

Question 1: Text Case Converter

Task: Ask the user to enter a sentence. Display the same sentence in several different letter cases.

Program requirements

1. Store the case-conversion method names in a collection and use a loop to display each result.
2. Display the text using `upper()`, `lower()`, `title()`, `capitalize()`, `swapcase()`
3. Use conditions with `isupper()`, `islower()`, and `istitle()` to describe the original text.

Example input: welcome to PYTHON

Upper	:	WELCOME TO PYTHON
Lower	:	welcome to python
Title	:	Welcome To Python

Capitalized : Welcome to python
Swap case : WELCOME TO python

Methods practised: upper(), lower(), title(),
capitalize(), swapcase(), casefold(), isupper(),
islower(), istitle()

Try this too: Test the program using PYTHON
IS FUN and Learning python.

Question 2: Username Validator

Task: Repeatedly ask the user for a username and report which validation rules it passes. Stop when the user enters quit.

Program requirements

1. Use a while loop to keep accepting usernames until quit is entered.
2. Use `isalnum()` to check whether the complete username contains only letters and numbers.
3. Use `isalpha()` on the first character to check whether the username begins with a letter.
4. Use `isidentifier()` to test whether it is a valid Python identifier, and `isascii()` to check its characters.
5. Use if and else statements to print clear validation messages.

Example input: student_10

Valid Python identifier

Contains only ASCII characters

Begins with a letter

Does not contain only letters and numbers

Methods practised: isalnum(), isalpha(),
isidentifier(), isascii()

Try this too: Test student10, 10student,
student_name, and विद्यार्थी.

Question 3: Formatted Student Report

Task: Collect the names and marks of three students, assign a grade, and display a neatly aligned report.

Grade conditions

1. 80 or above → A
2. 60 to 79 → B
3. 40 to 59 → C
4. Below 40 → Fail

Program requirements

1. Use a loop to collect three student names and marks.
2. Use if, elif, and else to determine each grade.
3. Use center() for the report heading, ljust() for names, and rjust() for marks and grades.
4. Use an f-string or format() to construct each output row.

STUDENT REPORT

Name	Marks	Grade
Asha	85	A
Rahul	63	B
John	35	Fail

Concepts practised: center(), ljust(), rjust(), f-strings, format(), loops, and if-elif-else

Try this too: Add a condition that rejects marks below 0 or above 100.

Question 4: Character and Text Analyser

Task: Read one line of text, examine each character, and report information about both the characters and the complete line.

Program requirements

1. Use a for loop to examine every character.
2. Use conditions to count letters, digits, spaces, printable characters, and non-printable characters.
3. Use `isalpha()`, `isdigit()`, `isspace()`, and `isprintable()` while counting.
4. After the loop, use `islower()`, `isupper()`, and `istitle()` to describe the complete text.
5. Display all totals using formatted output.

Example input: Python Class 101

```
Letters    : 11
Digits     : 3
```

```
Spaces      : 2  
Printable   : 16  
Title case  : True
```

Methods practised: `isalpha()`, `isdigit()`, `isspace()`, `isprintable()`, `islower()`, `isupper()`, `istitle()`

Try this too: Test an all-uppercase sentence and a line containing a tab character (`\t`).

Completion checklist

1. Every program must contain at least one loop.
2. Every program must contain at least one condition.
3. The output uses meaningful labels and is easy to read.
4. Each program has to be tested with more than one input.